

Experiment:1 Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

Aim: To simulate and analyze the electromagnetic field patterns of TEM, TE, and TM propagation modes using MATLAB.

Apparatus:

- Computer/Laptop
- MATLAB software

Procedure:

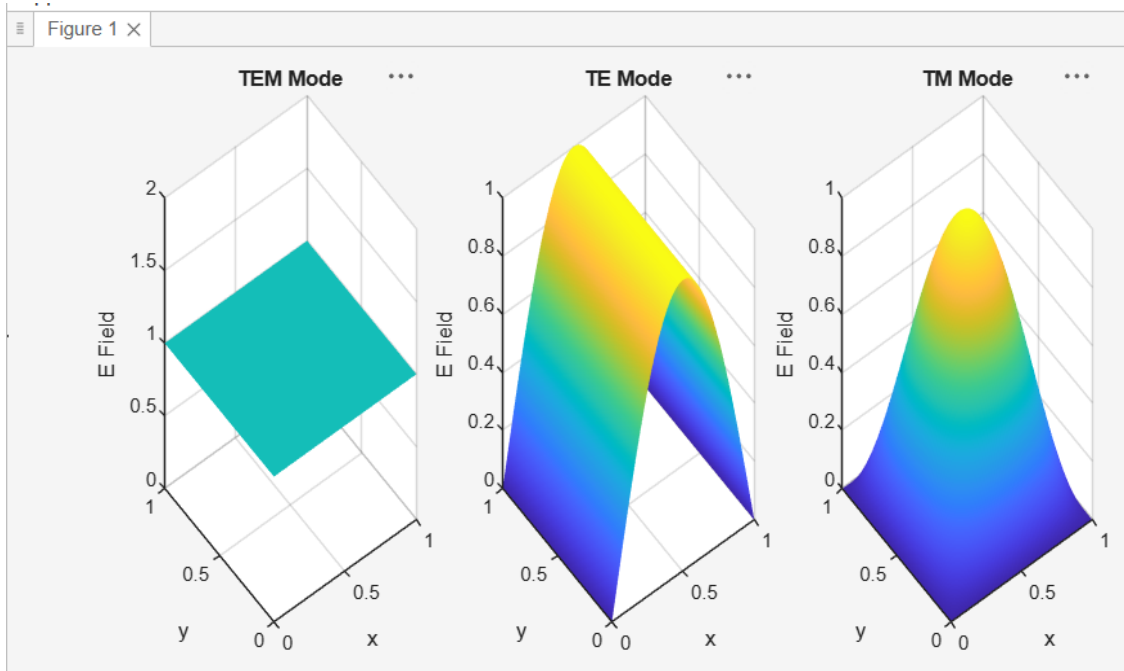
1. Open MATLAB and create a new script.
2. Define the x and y coordinates using linspace.
3. Create the coordinate grid using meshgrid.
4. Define the field patterns for TEM, TE, and TM modes.
5. Use subplot() to display all three modes in one figure.
6. Use surf() to plot the 3D electromagnetic field distributions.
7. Add suitable titles, axis labels, and field labels.
8. Run the MATLAB program and observe the three propagation modes.
9. Compare the field distributions of TEM, TE, and TM modes.
10. Record the obtained plots and result.

Code:

```
clc;
clear;
close all;
x = linspace(0,1,50);
y = linspace(0,1,50);
[X,Y] = meshgrid(x,y);
TEM = ones(size(X));
TE = sin(pi*X);
TM = sin(pi*X).*sin(pi*Y);
figure;
subplot(1,3,1);
surf(X,Y,TEM);
```

```
title('TEM Mode');  
xlabel('x');  
ylabel('y');  
zlabel('E Field');  
shading interp;  
subplot(1,3,2);  
surf(X,Y,TE);  
title('TE Mode');  
xlabel('x');  
ylabel('y');  
zlabel('E Field');  
shading interp;  
subplot(1,3,3);  
surf(X,Y,TM);  
title('TM Mode');  
xlabel('x');  
ylabel('y');  
zlabel('E Field');  
shading interp;
```

Output:



Result:

The field distributions of TEM, TE, and TM modes are successfully simulated and plotted using MATLAB.